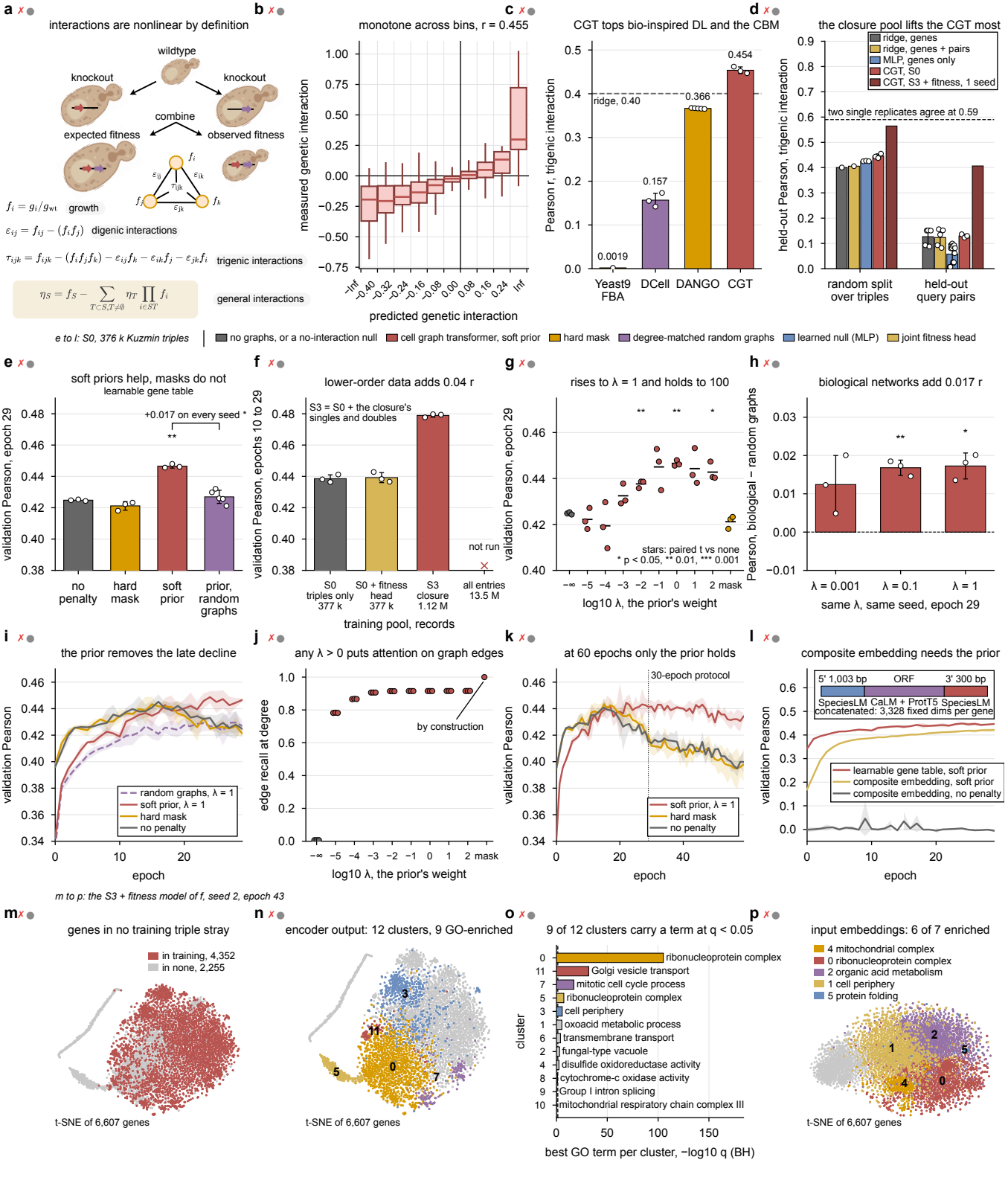

a
interactions are nonlinear by definition
wildtype
knockout
knockout
combine
expected fitness
observed fitness
$f_i = g_i/g_{wt}$ growth
$\varepsilon_{ij} = f_{ij} - (f_i f_j)$ digenic interactions
$\tau_{ijk} = f_{ijk} - (f_i f_j f_k) - \varepsilon_{ij} f_k - \varepsilon_{ik} f_j - \varepsilon_{jk} f_i$ trigenic interactions
$\eta_S = f_S - \sum_{T \subset S, T \neq \emptyset} \eta_T \prod_{i \in ST} f_i$ general interactions
b
monotone across bins, r = 0.455
measured genetic interaction
predicted genetic interaction
c
CGT tops bio-inspired DL and the CBM
Pearson r, trigenic interaction
ridge, 0.40
0.0019
0.157
0.366
0.454
Yeast9 FBA
DCell
DANGO
CGT
d
the closure pool lifts the CGT most
ridge, genes
ridge, genes + pairs
MLP, genes only
CGT, S0
CGT, S3 + fitness, 1 seed
held-out Pearson, trigenic interaction
two single replicates agree at 0.59
random split over triples
held-out query pairs
e to l: S0, 376 k Kuzmin triples
no graphs, or a no-interaction null
cell graph transformer, soft prior
hard mask
degree-matched random graphs
learned null (MLP)
joint fitness head
e
soft priors help, masks do not
learnable gene table
+0.017 on every seed *
**
validation Pearson, epoch 29
no penalty
hard mask
soft prior
prior, random graphs
f
lower-order data adds 0.04 r
S3 = S0 + the closure's singles and doubles
not run
validation Pearson, epochs 10 to 29
S0 triples only 377 k
S0 + fitness head 377 k
S3 closure 1.12 M
all entries 13.5 M
training pool, records
g
rises to λ = 1 and holds to 100
stars: paired t vs none
* p < 0.05, ** 0.01, *** 0.001
validation Pearson, epoch 29
log10 λ, the prior's weight
h
biological networks add 0.017 r
Pearson, biological − random graphs
λ = 0.001
λ = 0.1
λ = 1
same λ, same seed, epoch 29
i
the prior removes the late decline
random graphs, λ = 1
soft prior, λ = 1
hard mask
no penalty
validation Pearson
epoch
j
any λ > 0 puts attention on graph edges
by construction
edge recall at degree
log10 λ, the prior's weight
k
at 60 epochs only the prior holds
30-epoch protocol
soft prior, λ = 1
hard mask
no penalty
validation Pearson
epoch
l
composite embedding needs the prior
5' 1,003 bp
ORF
3' 300 bp
SpeciesLM
CaLM + ProtT5
SpeciesLM
concatenated: 3,328 fixed dims per gene
learnable gene table, soft prior
composite embedding, soft prior
composite embedding, no penalty
validation Pearson
epoch
m to p: the S3 + fitness model of f, seed 2, epoch 43
m
genes in no training triple stray
in training, 4,352
in none, 2,255
t-SNE of 6,607 genes
n
encoder output: 12 clusters, 9 GO-enriched
t-SNE of 6,607 genes
o
9 of 12 clusters carry a term at q < 0.05
cluster
ribonucleoprotein complex
Golgi vesicle transport
mitotic cell cycle process
ribonucleoprotein complex
cell periphery
oxoacid metabolic process
transmembrane transport
fungal-type vacuole
disulfide oxidoreductase activity
cytochrome-c oxidase activity
Group I intron splicing
mitochondrial respiratory chain complex III
best GO term per cluster, −log10 q (BH)
p
input embeddings: 6 of 7 enriched
4 mitochondrial complex
0 ribonucleoprotein complex
2 organic acid metabolism
1 cell periphery
5 protein folding
t-SNE of 6,607 genes